

1 Warm-start

F answers turns 1–W



large LLM



context-reply pairs

2 Soft-prompt mining

probe frozen G on turns 1–W



F's token and its neighbors

neighborhood
unlikelihood

expectation
weighting

entropy
regularizer

hardness of each turn



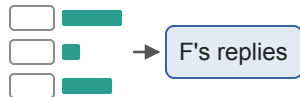
3 Localized LoRA

fit G to F's replies



frozen

on attention and MLP



turns weighted by hardness

adapted G



4 Gated serving

switch if both checks pass

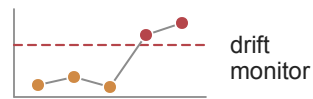


fidelity

locality



summary + last K turns



drift
monitor

t1

t2

t3

t4

t5

t6

t7

t8

one-time adaptation between turn 3 and turn 4 (steps 2–3)

drift detected at turn 8: roll back to F and refresh the warm-start state

trained

frozen